



Real Task Descriptions
(272 distinct tasks)



Real User Profiles
(5 dimensions)

Candidate Skill Repository
(228,432 unique skills)



...

Generate tasks and profiles



Skills



LLM



Tasks



Profiles

Structured Profiles

country/region
language
preferred platforms
payment methods
other constraints

Retrieve-then-Verify



Existing Pretrained Embedding Models
(Top-K Candidates)



LLM Scoring and Selection
(Six Dimensions)



1

Benchmark Quality Control



Low-overlap Filtering
(> 0.50 discarded)



Profile Counterfactual Augmentation
(Different profiles $\rightarrow$ different skills)

2



Verification and Human Arbitration
(to resolve disagreements)



Evaluation Dataset

(329 queries spanning 272 tasks, verified ground truth)



Low-overlap Filtering

Embedding Models

Semantic Hard Negative Mining

(Exclude True Positive from Top-K Similar Skills)

Task-centric Supervision Tuple (p, q, S^+, S^-)

Profile Counterfactual Augmentation

Give (q, p, S^+)

LLM

Re-select $\tilde{S}^+$

LLM generates a
counterfactual persona $\tilde{p}$

Mining New $\tilde{S}^-$

Counterfactual Supervision tuple $(\tilde{p}, q, \tilde{S}^+, \tilde{S}^-)$

Through Steps ① and ②

Profile-aware Training Dataset